

NATIONAL UNIVERSITY OF COMPUTER & EMERGING SCIENCES

Predictive Modeling and Micro-Climate Mitigation

Of Urban Islands in Karachi Using GEE and Machine Learning

Syed Waleed Hussain 23K-0885

Sofia Ayaz 23K-0807

Abdul Salam 23K-0806

Saniya 23K-0658

Ebrahim Moin 23K-0540

The Urban Heat Island (UHI) Crisis

The Global Challenge

Rapid urbanization replaces natural surfaces with heat-absorbing materials like concrete and asphalt. This leads to urban centers being significantly hotter than their rural surroundings.

Impact on Karachi

As a coastal megacity, Karachi faces intense thermal stress due to unregulated urban sprawl and recurring heatwaves. This study uses satellite data to map this crisis from 2018 to 2025.



Table Of Contents

1. Introduction
2. Literature Review
3. Methodology
4. Results
5. Discussion
6. Conclusion
7. Recommendations
8. References

The Urban Heat Island (UHI) Crisis

Problem Statement

1. Cities are getting hotter than surrounding rural areas.
2. Existing studies focus on past analysis only.
3. Lack of predictive and solution-based models

Research question

1. How can machine learning predict urban heat patterns?
2. how can mitigation strategies reduce temperature?



The Urban Heat Island (UHI) Crisis

Objectives

1. Analyse Land Surface Temperature (LST) from 2018-2025.
2. Study impact of NDVI, NDBI NDWI.
3. Develop ML model for prediction.
4. Suggest mitigation strategies

Scope

1. Karachi Metropolitan region (2018-2025 data).
2. Satellite based analysis .
3. Google Earth Engine (GEE) machine learning



The Urban Heat Island (UHI) Crisis

Limitations

1. Focus on surface temperature only.
2. Dependent on satellite data accuracy.
3. Simulation-based mitigation

Significance

1. Helps Urban Planners.
2. Supports climate resilience .
3. Moves from Observation
4. Prediction
5. Solution



Literature Review

Author	Problem	Method	Result	Conclusion
Almeida etl. (2021)	UHI monitering	Remote Sensing	Landsat effective	Satellite mapping essential
Elsevier (2024)	Karachi UHI growth	Geospatial analysis	Temp increased 2013-2023	No migitation simulation
Macarof & Statescu (2017)	NDVI vs NDBI	Landsat analysis	NDBI strongly linked	Built-up areas heat more
Roy & Bari (2022)	LST relationship	GEE platform	Vegetation reduces heat	GEE simplifies big data
Miniandi etal. (2025)	ML for UHI	Random Forest	ML improves accuracy	AI better than basic stats

Literature Review Summary



Vegetation vs. Concrete

Studies show a strong inverse correlation between NDVI (vegetation) and LST. While greenery provides cooling via evapotranspiration, impervious built-up areas (NDBI) act as heat traps.



Technological Gaps

Traditional research excels at historical mapping but lacks prognostic solutions. This research leverages Google Earth Engine (GEE) and AI to move from observation to simulation.

Terminology



NDVI

Normalized Difference Vegetation Index: Quantifies cooling green infrastructure and vegetation health.



NDBI

Normalized Difference Built-up Index: Measures heat-trapping constructed zones and urban density.



Albedo

Surface reflectivity: Essential for measuring how much solar radiation is absorbed vs. reflected by the city.



NDWI

A satellite-derived index used to detect water content or moisture in an area.



LST

The temperature of the Earth's surface as measured from satellite data.

Methodology



Population

Study Area : Karachi



Sample & Sampling Method

2018-2025 landsat 8 images
48,000 samples
30m resolution grid



Research Tools

Google Earth Engine
Landsat 8 (collection 2 level 2)
Python (MLP , RF, XGBoost,
SVR)



Feature Engineering

LST (Temperature)
NDVI (Vegetation)
NDBI (Built-up Areas)
NDWI (Moisture)
Albedo



Machine Learning

Model: XGBoost
Metric : RMSE, MSE, R^2
Train_test split : 80/20

Results (Findings)

1. High temperatures in central & industrial areas.
2. Coastal areas show lower temperatures
3. Clear spatial temperature variation

NDVI vs LST

1. High vegetation-> Low temperature
2. Low vegetation-> High temperature
3. Green areas help cooling

NDBI vs LST

1. High built-up areas -> High temperature
2. concrete surfaces retain heat
3. Strong positive correlation



Model Evaluation

Model	RMSE ↓	MAE ↓	R ² ↑
XGBoost	1.340	1.019	0.863
Random Forest	1.374	1.019	0.856
MLP (Neural Net)	1.525	1.178	0.823
SVR	1.803	1.338	0.752

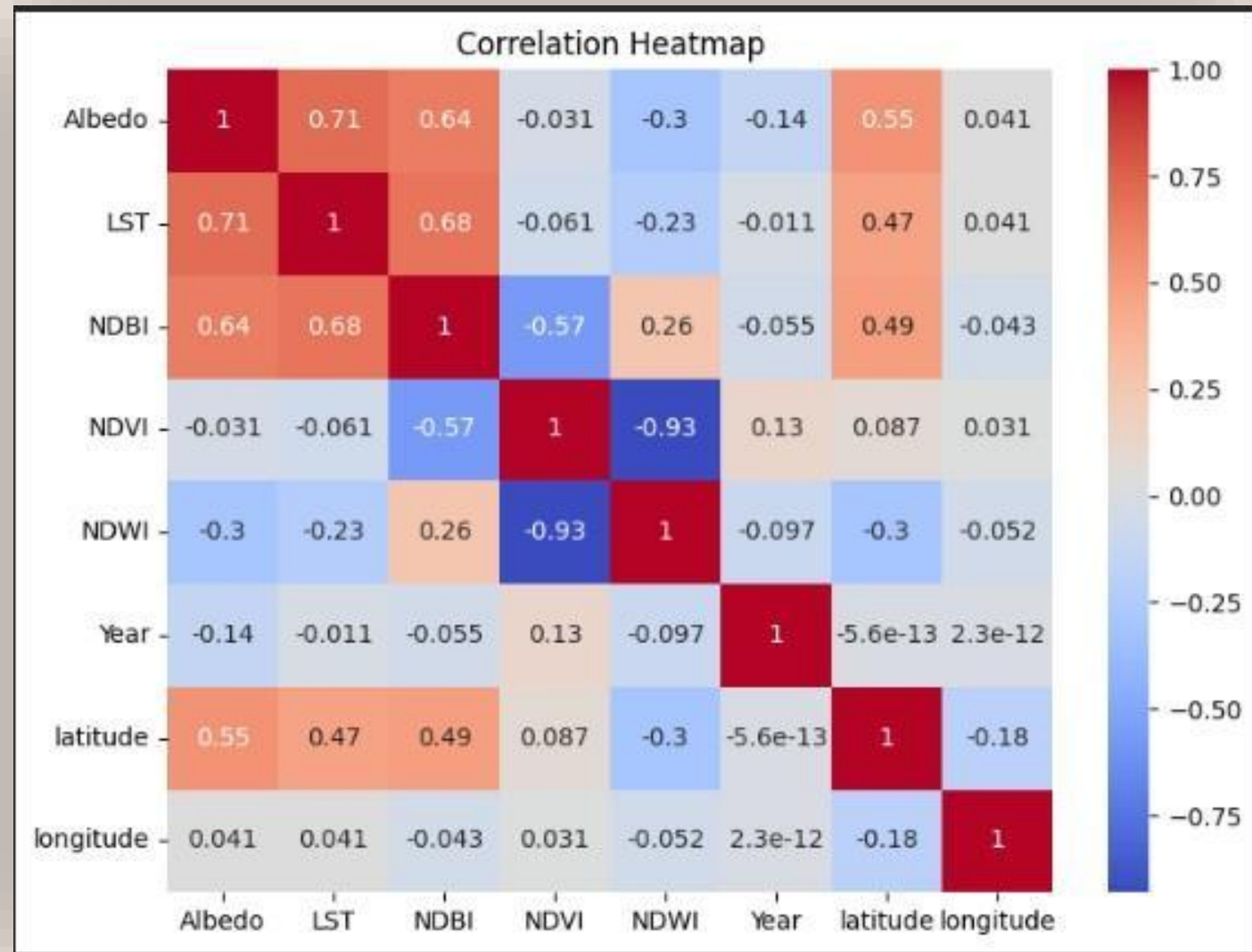
Metrics

- ✓ Lesser the RMSE, the better model performs.
- ✓ Lesser the MAE, the better model performs.
- ✓ Higher the R², the better model understand and explains the data.

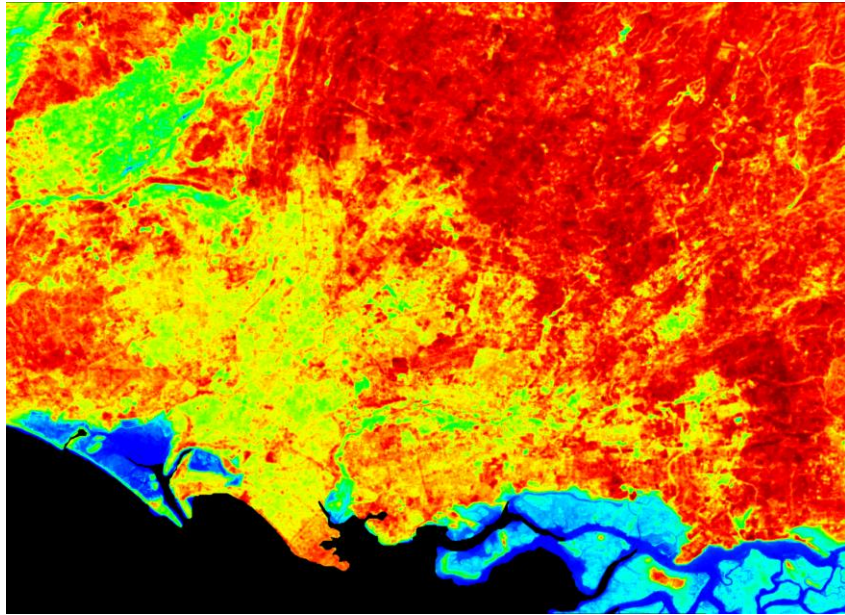
Best Selection

- ✓ Considering all the metrics, XGBoost model performs better than other ML models

Correlation Matrix

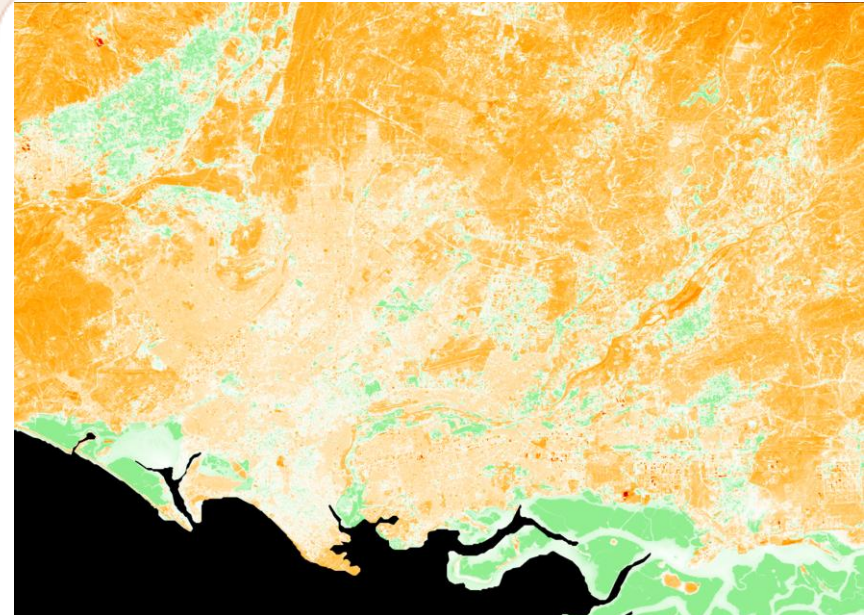


Spatial Mapping Findings



Thermal Hotspots

Intense heat in central and NW industrial districts.
And Cooler environment near the coastal areas.



Vegetation Deficit & Concrete Density

Low green and Orange cover corresponds with peak temperatures. High NDBI scores are the primary driver of UHI.

Discussion

1. Concrete surfaces are primary heat drivers
2. Vegetation significantly reduces heat
3. ML models outperform traditional regression
4. GEE allows scalable city-level analysis
5. Simulation enables proactive planning

Conclusion

1. Strong link between urban density and heat
2. Lack of greenery causes hotspots
3. ML model helps predict temperature trends
4. Framework supports better urban planning
5. 10% vegetation increase reduces heat up to 1.2 degree Celsius

Mitigation Recommendations



Strategic Urban Greening: Prioritize high-temperature central districts for green canopy initiatives to maximize cooling.



Adopt Albedo-Rich Materials: Mandate "cool roofs" and high-reflectivity materials for new urban infrastructure.

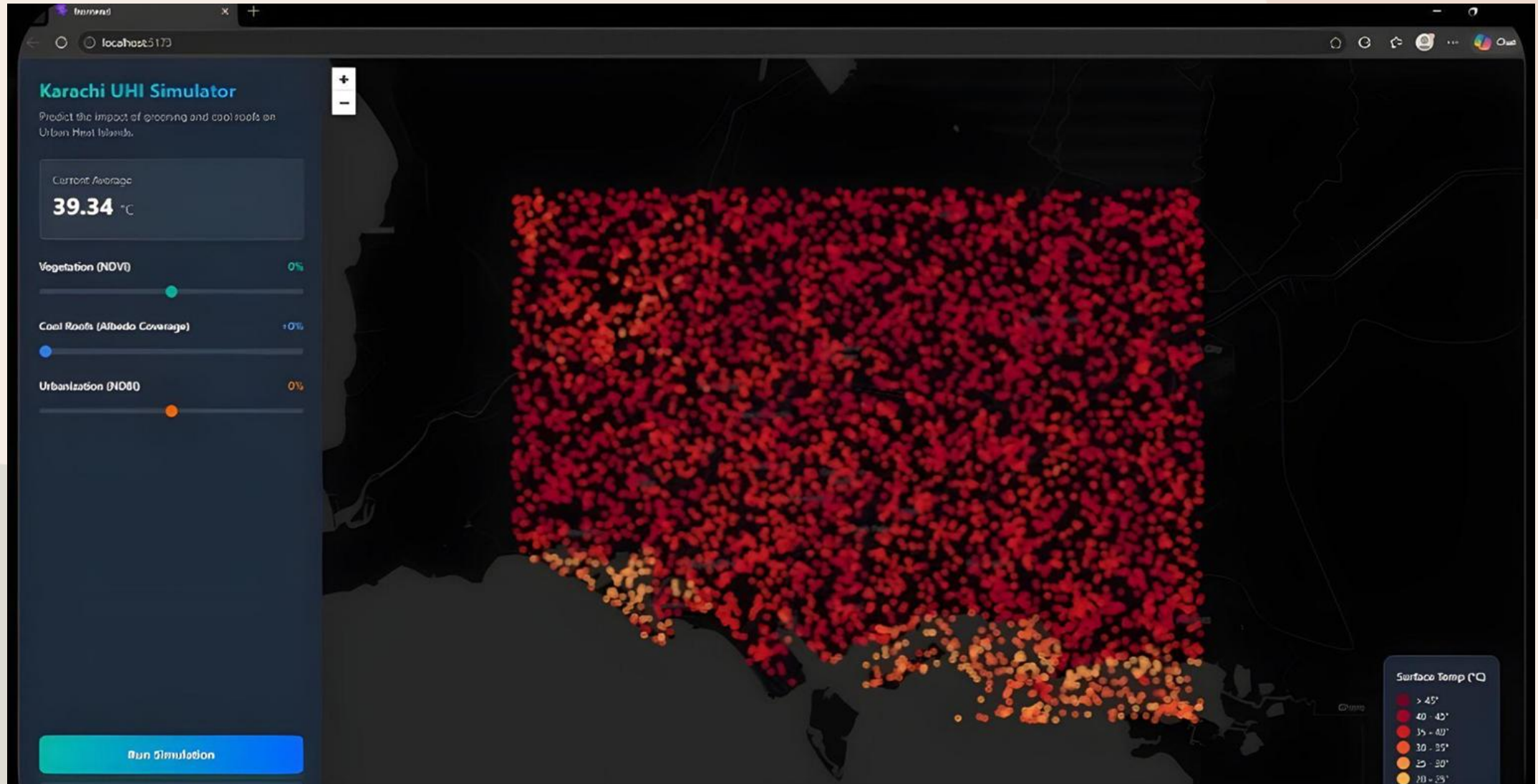


Climate-Responsive Planning: Restrict unchecked densification in climatically sensitive zones and protect ecological corridors.

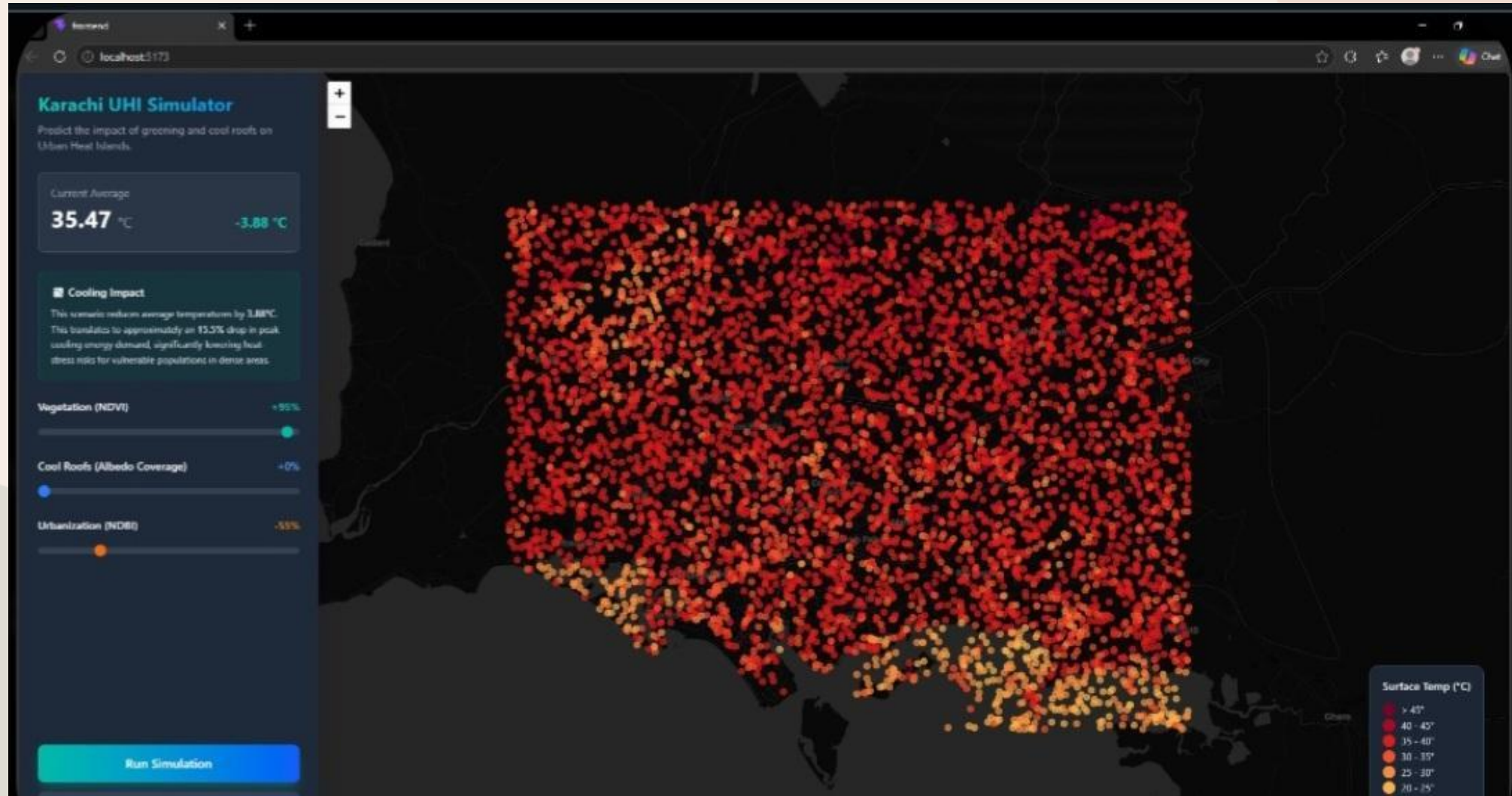


Real-Time Monitoring: Scale the GEE framework for seasonal thermal monitoring to protect vulnerable populations.

Simulation



Simulation



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